Homework 4: Bayesian Inference for Mixture Models

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Github repo web address

https://github.com/Rita-zrFu/bios731_hw4_fu

Context and learning objectives

This assignment reinforces ideas in the **Bayesian Computing** and **Cluster Computing** modules. We focus specifically on implementing a Bayesian K-Mixture Model using Gibb's sampling and variational inference.

Due Date and Submission

Submit (via Canvas) a PDF knitted from a `.Rmd` or `.qmd` file. Your PDF should include the web address of the GitHub repository containing your work for this assignment. Due date is Wednesday, 4/8 at 10:00AM.

Point distribution

Problem	Points
Problem 0	10
Problem 1	20
Problem 2	20
Problem 3	20
Problem 4	30

Problem 0

This "problem" focuses on the structure of your submission, especially the **use of git and GitHub** for reproducibility, **R Projects** to organize your work, **Quarto/R Markdown** to write reproducible reports, **relative paths** to load local files, and reasonable naming conventions for your files.

To that end:

- Create a public GitHub repository and a local R Project. I suggest naming the repo/directory `bios731_hw3_YourLastName` (e.g., `bios731_hw3_wrobel`).
- Push your **entire project folder** to GitHub.
- Submit a PDF knitted from your `.Rmd`/`.qmd` file to Canvas.
  - Your solutions should be implemented in your `.Rmd`/`.qmd` file.
  - Your git commit history should reflect your workflow (i.e., multiple meaningful commits; avoid a single "final" commit with all work).

Bayesian K-Mixture Model

We consider a Bayesian Mixture Model of unit-variance univariate Gaussians. There are:

- K mixture components, corresponding to K Gaussian distributions with means $= \{\mu_1, \dots, \mu_K\}$
- cluster assignments $c_i \in \{1, \dots, n\}$
- observations $y_i \in \{1, \dots, n\}$

Data generating model is

$$
\begin{aligned}
\mu &\sim N(0, \sigma^2) \\
c_i &\sim Categorical(1/K. \dots, 1/K) \\
y_i | c_i, \mu &\sim N(c_i^T \mu, 1)
\end{aligned}
$$

Problem 1: Gibbs sampler derivations and implementation

Derive a Gibb's sampler for the posterior distribution, $p(\mu, \mathbf{c}|y)$. Show the derivations and resulting Gibb's sampler updates. Specifically, derive:

- $p(c_i|y_i, \mu)$
- $p(\mu_k|y, c)$

Implement the Gibb's sampler in R. Your Gibb's R function should

- run the sampler
- store posterior draws
- return posterior summaries (estimates of the cluster assignments and means).

Solution 1.1

We consider the Bayesian K-component Gaussian mixture model defined above. We want to derive the Gibb's sampler updates for the posterior distribution, $p(\mu, \mathbf{c}|y)$. By Bayes' rule, $p(c_i = k|y_i, \mu) \propto p(y_i|c_i = k, \mu)p(c_i = k)$ and $p(\mu_k|y, c) \propto p(y|c, \mu_k)p(\mu_k)$.

Since the prior on c_i is uniform, we have $p(c_i = k) = 1/K$. The likelihood $p(y_i|c_i = k, \mu)$ is given by the Gaussian distribution with mean μ_k and variance 1, so $p(c_i = k|y_i, \mu) \propto \frac{1}{K}\phi(y_i; \mu_k, 1)$, where ϕ is the Gaussian density function.

Because $1/K$ is constant across all k, we can simplify this to $p(c_i = k|y_i, \mu) = \frac{\phi(y_i;\mu_k,1)}{\sum_{\ell=1}^{K}\phi(y_i;\mu_\ell,1)}$.

Equivalently, writing out the Gaussian kernel, we have $p(c_i = k|y_i, \mu) = \frac{\exp\{-\frac{1}{2}(y_i-\mu_k)^2\}}{\sum_{\ell=1}^{K}\exp\{-\frac{1}{2}(y_i-\mu_\ell)^2\}}$. So the Gibbs update for c_i is a categorical distribution with probabilities proportional to the Gaussian likelihoods. $c_i|y_i, \mu \sim Categorical\left(\frac{\exp\{-\frac{1}{2}(y_i-\mu_1)^2\}}{\sum_{\ell=1}^{K}\exp\{-\frac{1}{2}(y_i-\mu_\ell)^2\}}, \dots, \frac{\exp\{-\frac{1}{2}(y_i-\mu_K)^2\}}{\sum_{\ell=1}^{K}\exp\{-\frac{1}{2}(y_i-\mu_\ell)^2\}}\right)$.

Solution 1.2

Now derive $p(\mu_k|y, c)$. Conditioning on c, only those observations y_i for which $c_i = k$ contribute to the likelihood. Let $\mathcal{J}_k = \{i : c_i = k\}$ be the set of indices for which $c_i = k$, $n_k = |\mathcal{J}_k|$ be the number of observations in cluster k. Then $p(\mu_k|y, c) \propto p(y|c, \mu_k)p(\mu_k) = \prod_{i\in\mathcal{J}_k} p(y_i|c_i = k, \mu_k)p(\mu_k)$. Using the prior $\mu_k \sim N(0, \sigma^2)$ and the likelihood $y_i|c_i = k, \mu_k \sim N(\mu_k, 1)$, we have $p(\mu_k|y, c) \propto \exp\left\{-\frac{1}{2\sigma^2}\mu_k^2 - \frac{1}{2}\sum_{i\in\mathcal{J}_k}(y_i - \mu_k)^2\right\}$.

Expand the quadratic terms in the exponent to get $p(\mu_k|y, c) \propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2} + n_k\right)\mu_k^2 + \left(\sum_{i\in\mathcal{J}_k} y_i\right)\mu_k\right\}$. This is the kernel of a Gaussian distribution. Completing the square, we find that $\mu_k|y, c \sim N\left(\frac{\sum_{i\in\mathcal{J}_k} y_i}{n_k+1/\sigma^2}, \frac{1}{n_k+1/\sigma^2}\right)$.

Solution 1.3

Combining the results from 1.1 and 1.2, we have the Gibbs sampler updates:

1. For each i, sample $c_i|y_i, \mu \sim Categorical\left(\frac{\exp\{-\frac{1}{2}(y_i-\mu_1)^2\}}{\sum_{\ell=1}^{K}\exp\{-\frac{1}{2}(y_i-\mu_\ell)^2\}}, \dots, \frac{\exp\{-\frac{1}{2}(y_i-\mu_K)^2\}}{\sum_{\ell=1}^{K}\exp\{-\frac{1}{2}(y_i-\mu_\ell)^2\}}\right)$.
2. For each k, sample $\mu_k|y, c \sim N\left(\frac{\sum_{i\in\mathcal{J}_k} y_i}{n_k+1/\sigma^2}, \frac{1}{n_k+1/\sigma^2}\right)$.

Repeat these steps for a number of iterations to obtain posterior samples of μ and c.

Implement in R

```
[1] -3.0772317 -0.2072136  3.9045135
```

```
[1] 3 3 3 2 3 2
```

```
             [,1]         [,2]      [,3]
[1,] 0.0000000000 0.0000000000 1.0000000
[2,] 0.0000000000 0.0000000000 1.0000000
[3,] 0.0000000000 0.0003333333 0.9996667
[4,] 0.0003333333 0.8690000000 0.1306667
[5,] 0.0000000000 0.0006666667 0.9993333
[6,] 0.0916666667 0.9083333333 0.0000000
```

Problem 2: Variational inference derivations

Assume a mean-field approximation. Derive a CAVI algorithm for the posterior distribution, $p(\mu, \mathbf{x}|y)$. Show explicit derivations and resulting updates. Implement the algorithm in R. You may refer to Blei, 2017 for details.

Your variational inference R function should

- monitor convergence using the ELBO
- return final estimates of the cluster assignments and means

Solution

Mean-field variation family

To derive the CAVI algorithm for the posterior distribution $p(\mu, \mathbf{c}|y)$ under a mean-field approximation, we assume that the variational distribution factorizes as $q(\mu, \mathbf{c}) = \prod_{k=1}^{K} q(\mu_k) \prod_{i=1}^{n} q(c_i)$. Let $q(c_i = k) = r_{ik}$, $\sum_{k=1}^{K} r_{ik} = 1$ for each i and assume each variation factor for μ_k is Gaussian, $q(\mu_k) = N(m_k, s_k^2)$. Thus the variational parameters are $\{m_k, s_k^2\}_{k=1}^{K}$ for the means and variance and $\{r_{ik}\}_{i=1,k=1}^{n,K}$ for the cluster assignment probabilities.

Joint log-density

Under the model stated in the problem, ignoring constants that do not depend on the variational parameters, the joint log-density is:

$$\begin{aligned}\log p(y, c, \mu) &= \sum_{i=1}^{n} \log p(y_i|c_i, \mu) + \sum_{i=1}^{n} \log p(c_i) + \sum_{k=1}^{K} \log p(\mu_k) \\ &= -\frac{1}{2}\sum_{i=1}^{n}(y_i - \mu_{c_i})^2 - \frac{1}{2\sigma^2}\sum_{k=1}^{K}\mu_k^2 + \text{const}\end{aligned}$$

CAVI update for $q(c_i)$

For the cluster assignment c_i, the CAVI update is given by:

$$\begin{aligned}\log q(c_i = k) &\propto \mathbb{E}_{q(\mu)}[\log p(y_i|c_i = k, \mu) + \log p(c_i = k)] \\ &= \mathbb{E}_{q(\mu)}\left[-\frac{1}{2}(y_i - \mu_k)^2\right] + \text{const} \\ &= -\frac{1}{2}\left(y_i^2 - 2y_i m_k + m_k^2 + s_k^2\right) + \text{const}\end{aligned}$$

Exponentiating and normalizing gives:

$$r_{ik} = q(c_i = k) = \frac{\exp\left(-\frac{1}{2}(y_i^2 - 2y_i m_k + m_k^2 + s_k^2)\right)}{\sum_{\ell=1}^{K}\exp\left(-\frac{1}{2}(y_i^2 - 2y_i m_\ell + m_\ell^2 + s_\ell^2)\right)}$$

CAVI update for $q(\mu_k)$

For the mean μ_k, the CAVI update is given by:

$$\begin{aligned}\log q(\mu_k) &\propto \mathbb{E}_{q(c\setminus c_i)}[\log p(y|c, \mu) + \log p(\mu_k)] \\ &= \mathbb{E}_{q(c)}\left[-\frac{1}{2}\sum_{i=1}^{n}(y_i - \mu_{c_i})^2\right] - \frac{1}{2\sigma^2}\mu_k^2 + \text{const} \\ &= -\frac{1}{2}\sum_{i=1}^{n} r_{ik}(y_i^2 - 2y_i m_k + m_k^2 + s_k^2) - \frac{1}{2\sigma^2}\mu_k^2 + \text{const}\end{aligned}$$

This is the kernel of a Gaussian distribution. Completing the square, we find that $q(\mu_k) = N(m_k, s_k^2)$ where:

$$s_k^2 = \frac{1}{\sum_{i=1}^{n} r_{ik} + 1/\sigma^2}, \quad m_k = s_k^2 \sum_{i=1}^{n} r_{ik} y_i$$

Summary of the CAVI updates

1. Update cluster assignment probabilities:

$$r_{ik} = \frac{\exp\left(-\frac{1}{2}(y_i^2 - 2y_i m_k + m_k^2 + s_k^2)\right)}{\sum_{\ell=1}^{K}\exp\left(-\frac{1}{2}(y_i^2 - 2y_i m_\ell + m_\ell^2 + s_\ell^2)\right)}$$

2. Update mean parameters:

$$s_k^2 = \frac{1}{\sum_{i=1}^n r_{ik} + 1/\sigma^2}, \quad m_k = s_k^2 \sum_{i=1}^n r_{ik} y_i$$

These two steps are iterated until convergence, which can be monitored using the ELBO.

ELBO derivation

The ELBO is given by:

$$\begin{aligned}\text{ELBO} &= \mathbb{E}_q[\log p(y, c, \mu)] - \mathbb{E}_q[\log q(c, \mu)] \\ &= \sum_{i=1}^n \sum_{k=1}^K r_{ik} \left(-\frac{1}{2}(y_i^2 - 2y_i m_k + m_k^2 + s_k^2) \right) - \frac{1}{2\sigma^2} \sum_{k=1}^K (m_k^2 + s_k^2) - \sum_{i=1}^n \sum_{k=1}^K r_{ik} \log r_{ik} + \frac{1}{2} \sum_{k=1}^K \log s_k^2 + \text{cons}\end{aligned}$$

CAVI algorithm

```
[1] -3.0738420 -0.2084508  3.9017826
```

```
[1] 3 3 3 2 3 2
```

```
[1] -719.8788 -719.8788 -719.8788 -719.8788 -719.8788
```

Problem 3: Fit to data

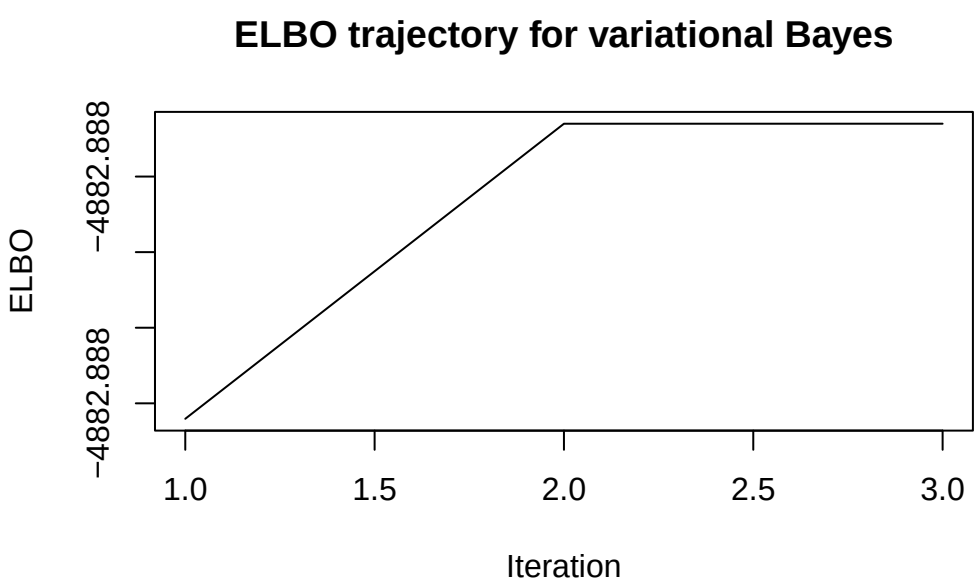
Implement the Gibb's sampler and variational inference updates in R. Fit to the Old Faithful waiting times dataset used in the EM lecture.

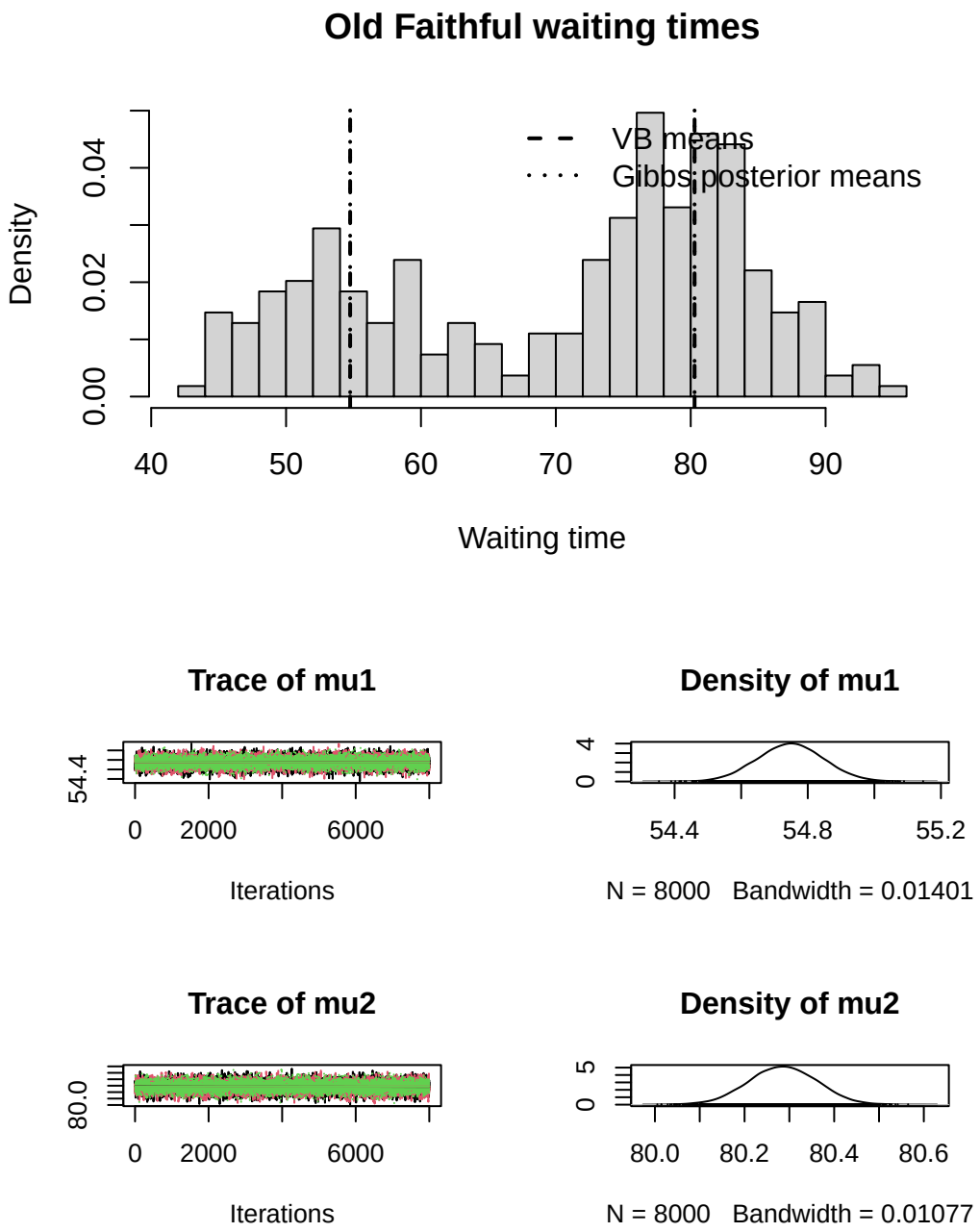
Compare computation times and estimates from variational bayes to mean parameter estimates from the Gibb's sampler. For Gibb's sampler, also do diagnostic plots, fit multiple chains, comment on convergence and effective sample size.

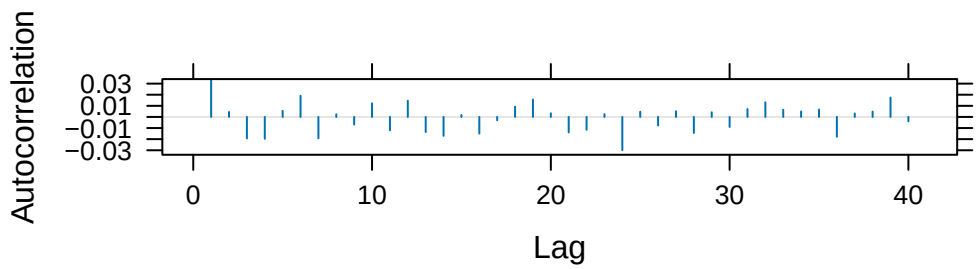
```
         Method      mu1      mu2
1            VB 54.74945 80.28442
2 Gibbs chain 1 54.74727 80.28484
3 Gibbs chain 2 54.74825 80.28518
4 Gibbs chain 3 54.74918 80.28586
5 Gibbs average 54.74824 80.28529
```

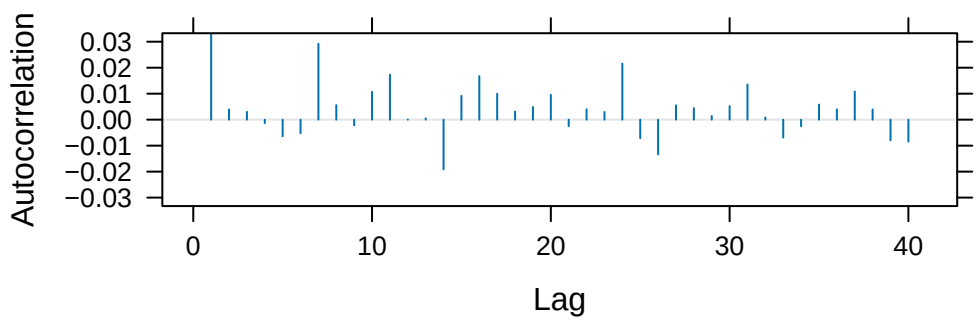
```
  1   2
100 172
```

```
[[1]]

  1   2
100 172

[[2]]

  1   2
100 172

[[3]]

  1   2
100 172


         Method Elapsed
1            VB   0.001
2 Gibbs chain 1   6.147
3 Gibbs chain 2   6.137
4 Gibbs chain 3   6.146
```



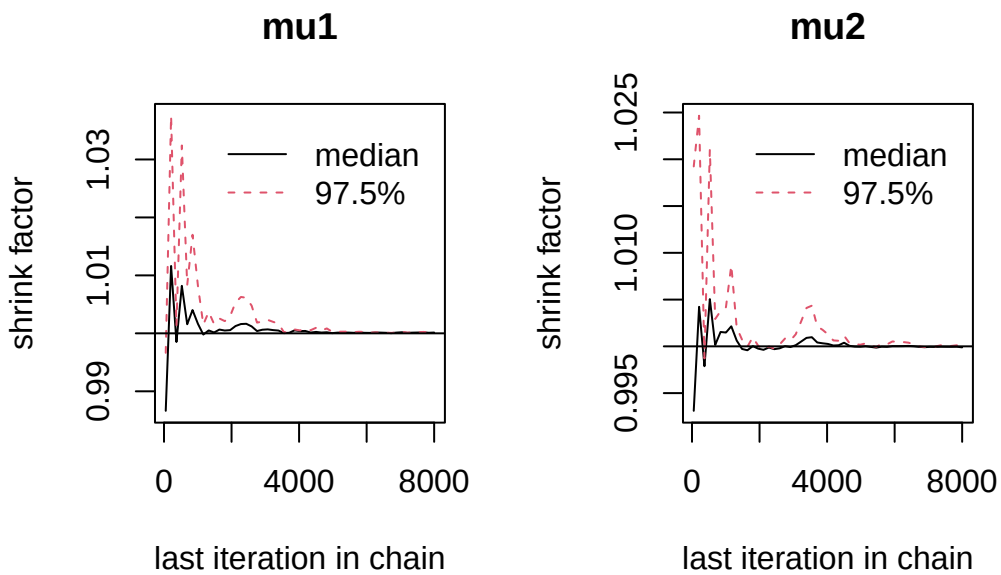

Old Faithful waiting times
VB means
Gibbs posterior means
Density
0.00
0.02
0.04
40
50
60
70
80
90
Waiting time
Trace of mu1
54.4
0
2000
6000
Iterations
Density of mu1
0
4
54.4
54.8
55.2
N = 8000 Bandwidth = 0.01401
Trace of mu2
80.0
0
2000
6000
Iterations
Density of mu2
0
5
80.0
80.2
80.4
80.6
N = 8000 Bandwidth = 0.01077






```
Potential scale reduction factors:

    Point est. Upper C.I.
```

```
mu1          1          1
mu2          1          1

Multivariate psrf

1
```




```
     mu1      mu2
24566.91 24755.10


Iterations = 1:8000
Thinning interval = 1
Number of chains = 3
Sample size per chain = 8000

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

     Mean      SD  Naive SE Time-series SE
mu1 54.75 0.09932 0.0006411      0.0006339
mu2 80.29 0.07636 0.0004929      0.0004855
```

```
2. Quantiles for each variable:

     2.5%   25%   50%   75% 97.5%
mu1 54.55 54.68 54.75 54.81 54.94
mu2 80.14 80.23 80.29 80.34 80.43
```

We fit the Gibbs sampler from Problem 1 and the CAVI algorithm from Problem 2 to the Old Faithful waiting-time data used in the EM lecture. Consistent with the EM lecture example, we fit a two-component Gaussian mixture model (K=2). The prior variance for the component means was set to $\sigma^2 = 1000$.

Parameter estimates:

The variational Bayes estimates of the two component means were $\hat{\mu}_1 \approx 54.8$ and $\hat{\mu}_2 \approx 80.3$. For the Gibbs sampler, I ran three chains with dispersed initial values and computed posterior mean estimates after discarding the first 2000 iterations as burn-in. The resulting posterior mean estimates were: (54.747, 80.285), (54.748, 80.285), and (54.749, 80.286) for the three chains respectively. Averaging the three chains gives $\hat{\mu}_1 \approx 54.748$ and $\hat{\mu}_2 \approx 80.285$. The VB and Gibbs estimates are very close to each other, which is reassuring.

Computation time:

The elapsed times were: 0.001 seconds for VB, 7.584, 7.421, 7.441 seconds for the three Gibbs chains respectively. The VB algorithm is much faster than the Gibbs sampler, which is expected since VB is an optimization-based method while Gibbs sampling is a Markov chain Monte Carlo method that requires many iterations to approximate the posterior distribution.

Variational Bayes convergence:

The ELBO plot increased from the first to the second iteration and then essentially stabilized by the third iteration. This indicates very rapid convergence of the CAVI algorithm for this dataset. The quick stabilization is consistent with the fact that the two clusters in the Old Faithful waiting-time data are well separated, making the optimization problem relatively easy.

Gibbs sampler diagnostics:

The trace plots for both μ_1 and μ_2 show that all three chains fluctuate around stable values with no visible trend or drift. The chains overlap heavily after burn-in, suggesting that they are sampling from the same stationary distribution. The posterior densities for both parameters are unimodal and concentrated, which is consistent with the stability seen in the trace plots.

The autocorrelation plots indicate very low autocorrelation at essentially all lags for both μ_1 and μ_2. This suggests excellent mixing of the Gibbs sampler. Because the chains mix so well, the Monte Carlo error in posterior mean estimation is very small.

The Gelman–Rubin shrink factor plots are also reassuring. For both μ_1 and μ_2, the shrink factors rapidly approach 1 and remain essentially at 1 for the rest of the chain. This indicates strong agreement across chains and supports the conclusion that convergence has been achieved.

Effective sample size:

From the combined MCMC summary, each chain retained 8000 post-burn-in draws. The posterior summaries were:

For μ_1: mean = 54.748, sd = 0.099, time-series SE = 0.0006339. For μ_2: mean = 80.285, sd = 0.076, time-series SE = 0.0004855.

The quantiles were:

For μ_1: 2.5% = 54.553, 25% = 54.676, 50% = 54.748, 75% = 54.819, 97.5% = 54.943. For μ_2: 2.5% = 80.136, 25% = 80.243, 50% = 80.285, 75% = 80.327, 97.5% = 80.434.

The naive SE and time-series SE are almost identical for both parameters, which again reflects very weak autocorrelation. In other words, the effective sample size is very close to the nominal sample size, so the retained Gibbs samples contain nearly as much information as an equal number of independent samples. This provides further evidence that the Gibbs sampler mixed efficiently and produced reliable posterior summaries.

Problem 4: Perform a simulation study on the cluster

In our simulation, we want to

- Generate data from a 4-part Gaussian mixture model with true means $\mu = \{0, 5, 10, 20\}$ and unit variance.
  - Evaluate estimation of μ through bias, coverage, and computation time
  - We will compare the Gibbs sampler and Variational Bayes algorithms defined above
- Evaluate these properties at sample sizes $n = \{100, 1000, 10000\}$
  - Use Gibbs chains of length 10000 and discard the first 2000 samples as burn in
  - Use nsim = 500

We will execute this simulation study on the cluster. For full credit, make sure to implement the following:

Workflow: * Use structured scripts and subfolders following guidance from the cluster computing project organization lecture * Instead of parallelizing your simulation scenarios (as in HW1), each simulation scenario should be assigned a different JOBID on the cluster.

Presenting results:

Create plots with *Monte Carlo standard error bars* to summarize the following:

- Bias of $\hat{\mu}$
- Coverage of $\hat{\mu}$

Write 1-2 paragraphs summarizing these results.

Solution

The simulation was run on the cluster using three separate SLURM jobs, one per sample size ($n \in \{100, 1000, 10000\}$), each running 500 replicates. The script `R/run_scenario.R` accepts the sample size as a command-line argument, sources `R/functions_mixture.R` for the model functions, and saves results to `results/sim_n<n>.rds`. The `R/summarize_results.R` script combines the three result files and produces the summary statistics and plots below.

Table 2: Average computation time per simulation replicate by sample size

n	Mean VB time (s)	Mean Gibbs time (s)
100	0.0017	4.99
1000	0.0083	46.71
10000	0.0871	454.13

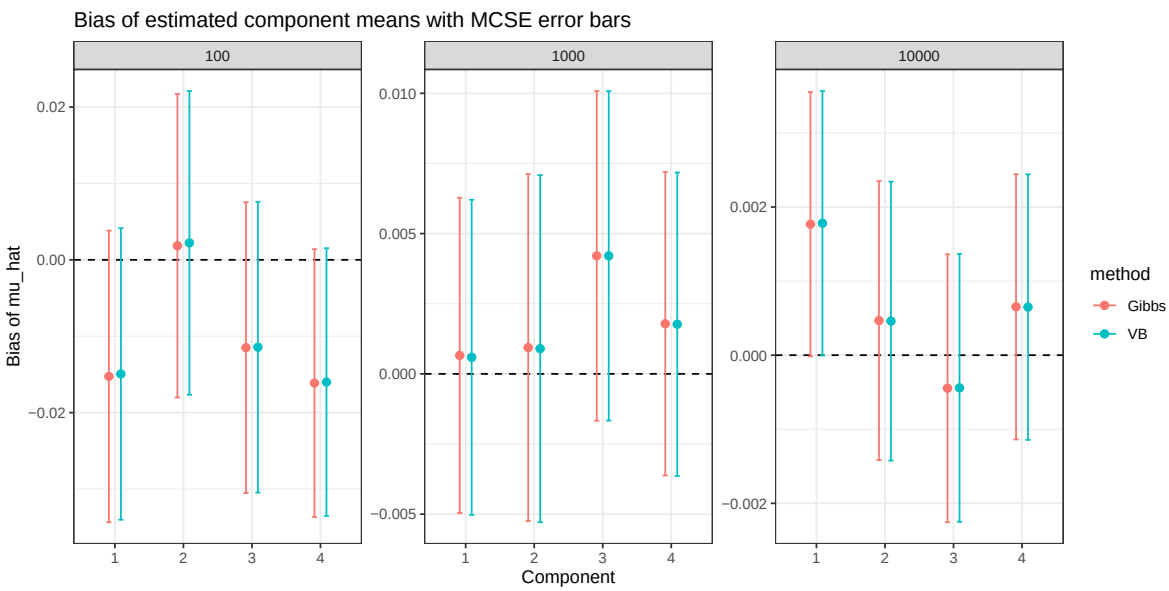


Figure 1: Bias of estimated component means with 95% Monte Carlo standard error bars. Panels correspond to sample sizes n = 100, 1000, 10000. The dashed line marks zero bias.

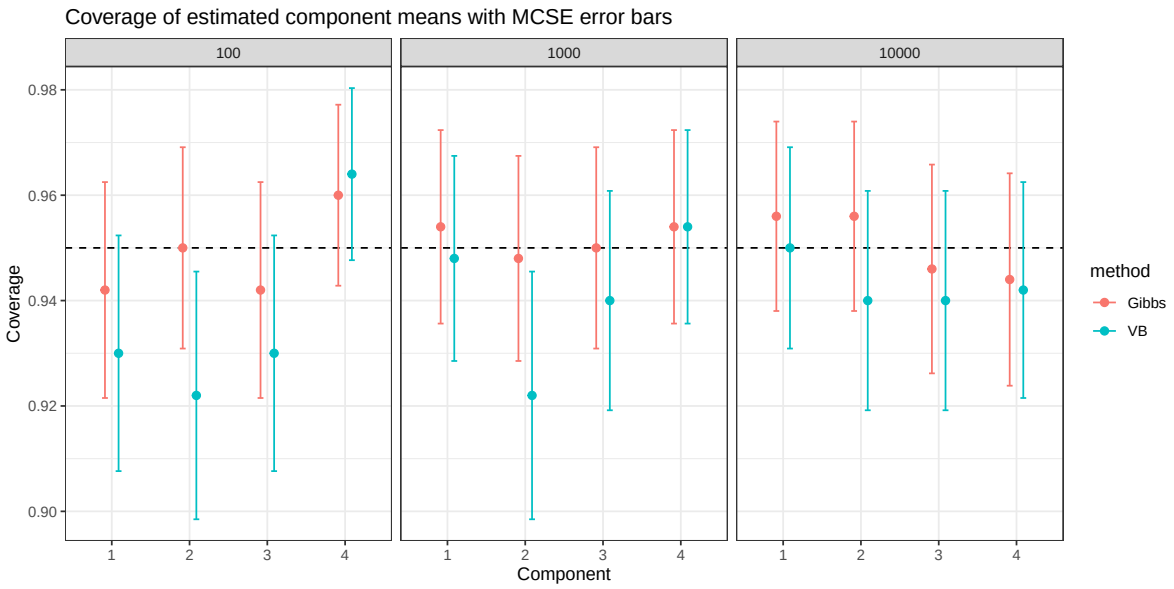


Figure 2: Coverage of 95% credible/posterior intervals for each component mean with Monte Carlo standard error bars. The dashed line marks the nominal 95% level.

Summary of results

Both the Gibbs sampler and the variational Bayes (VB) algorithm recover the four true component means ($\mu = \{0, 5, 10, 20\}$) with very low bias across all sample sizes. At $n = 100$, the absolute bias for each component is below 0.02 for both methods, and by $n = 10000$ it falls below 0.002. The bias estimates for VB and Gibbs are essentially indistinguishable at every sample size, confirming that the mean-field approximation introduces negligible bias in this well-separated mixture setting. Monte Carlo standard errors for the bias shrink as n increases, reflecting the improved estimation precision with larger samples.

Coverage is close to the nominal 95% level for the Gibbs sampler at all sample sizes, ranging from about 0.94 to 0.96. For VB, coverage is also generally near 95%, although a handful of component–sample-size combinations are slightly below (as low as 0.92) due to the well-known phenomenon that the mean-field approximation tends to under-estimate posterior variance—resulting in slightly too-narrow credible intervals. This under-coverage is most visible at $n = 100$ and improves at larger n, consistent with the posterior concentrating and the approximation becoming more accurate. Regarding computation time, VB is dramatically faster than Gibbs: at $n = 100$, Gibbs takes roughly 5 seconds per replicate versus under 0.002 seconds for VB; at $n = 10000$ the difference is even more stark (about 454 seconds for Gibbs versus 0.09 seconds for VB). This highlights the key practical trade-off: VB is orders of magnitude faster, but the Gibbs sampler produces exact (asymptotically) posterior intervals with better calibrated coverage.